

Time Series Exercise Sheet 12

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Exercise 12.1

Assume the generation of r_t is governed by an ARCH (AutoRegressive Conditionally Heteroscedastic) model. This particular instance takes the form for β_j with $j = 0, 1, 2$ non-negative constants and ε_t a zero-mean process:

$$\begin{aligned}r_t &= \sigma_t \varepsilon_t \\ \sigma_t^2 &= \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2,\end{aligned}$$

and ε_t is assumed to have unit variance for identifiability. Let F_t denote all the information available at time point t

- (a) Calculate $\mathbb{E}[r_t]$ using the law of iterated expectation, conditioning on F_{t-1} .
- (b) Assuming r_t is stationary, calculate the marginal variance of r_t , quoting any laws of probability you may need.
- (c) Determine necessary conditions for r_t to be stationary in terms of β_j with $j = 0, 1, 2$.

Solution 12.1

1. Since ε_t is independent of F_{t-1} ,

$$\begin{aligned}\mathbb{E}(r_t \mid F_{t-1}) &= \mathbb{E}(\sigma_t \varepsilon_t \mid F_{t-1}) \\ &= \sigma_t \mathbb{E}(\varepsilon_t) \\ &= 0 \\ \mathbb{E}(r_t) &= \mathbb{E}\{\mathbb{E}(r_t \mid F_{t-1})\} \\ &= 0\end{aligned}$$

2. Since ε_t is independent of F_{t-1} , by the law of total variance, it holds that

$$\begin{aligned}\text{Var}(r_t \mid F_{t-1}) &= \text{Var}(\sigma_t \varepsilon_t \mid F_{t-1}) \\ &= (\beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2) \text{Var}(\varepsilon_t) \\ &= \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2\end{aligned}$$

$$\begin{aligned}\text{Var}(r_t) &= \mathbb{E}[\text{Var}(r_t \mid F_{t-1})] + \text{Var}(\mathbb{E}[r_t \mid F_{t-1}]) \\ &= \mathbb{E}[\beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2] + 0 \\ &= \beta_0 + \beta_1 \text{Var}(r_{t-1}) + \beta_2 \text{Var}(r_{t-2})\end{aligned}$$

since $\mathbb{E}[r_t] = 0$ by (a). Then, $\text{Var}(r_t) = \text{Var}(r_{t-1}) = \dots = \text{Var}(r_1)$, it holds that

$$\begin{aligned}(1 - \beta_1 - \beta_2) \text{Var}(r_t) &= \beta_0 \\ \Rightarrow \text{Var}(r_t) &= \frac{\beta_0}{1 - \beta_1 - \beta_2}\end{aligned}$$

if $\beta_1 + \beta_2 < 1$.

3. To be stationary, $\beta_1 + \beta_2 < 1$, so the variance of r_t in (b) is positive.

Exercise 12.2

The generalised autoregressive conditionally heteroscedastic (GARCH) class of models assuming Gaussian noise is given by

$$r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \sum_{j=1}^m \alpha_j r_{t-j}^2 + \sum_{j=1}^r \beta_j \sigma_{t-j}^2, \quad \varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, 1) \quad (1)$$

and labelled GARCH(m, r). To avoid negative variances, we take $\alpha_j, \beta_j \geq 0$.

A GARCH($m, 0$) is an ARCH(m). Prove that a GARCH model has zero mean, and satisfies: $\text{Corr}(r_{t+\tau}, r_t) = 0$.

Remember that $\mathcal{F}_t$ denotes the information available up to time t .

Solution 12.2

Since σ_t is determined by the past, it is $\mathcal{F}_{t-1}$ -measurable. Therefore

$$\begin{aligned} \mathbb{E}[r_t] &= \mathbb{E}[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]] \\ &= \mathbb{E}[\mathbb{E}[\sigma_t \varepsilon_t \mid \mathcal{F}_{t-1}]] \\ &= \mathbb{E}[\sigma_t \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}]] \\ &= 0. \end{aligned}$$

Now let $\tau > 0$. Then

$$\begin{aligned} \text{Cov}(r_{t+\tau}, r_t) &= \mathbb{E}[r_{t+\tau} r_t] - \mathbb{E}[r_{t+\tau}] \mathbb{E}[r_t] \\ &= \mathbb{E}[r_{t+\tau} r_t] \\ &= \mathbb{E}[\mathbb{E}[r_{t+\tau} r_t \mid \mathcal{F}_{t+\tau-1}]] \\ &= \mathbb{E}[\mathbb{E}[\sigma_{t+\tau} \varepsilon_{t+\tau} r_t \mid \mathcal{F}_{t+\tau-1}]] \\ &= \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\varepsilon_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]] \\ &= 0. \end{aligned}$$

Therefore $\text{Corr}(r_{t+\tau}, r_t) = 0$.

Exercise 12.3

Show that for an ARCH(1) with parameters $0 < \alpha_0$, $0 < \alpha_1 < 1/3$ we have $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$

Solution 12.3

We start with the covariance, assuming that $\tau > 0$.

$$\begin{aligned} \text{Cov}(r_{t+\tau}^2, r_t^2) &= \mathbb{E}[r_{t+\tau}^2 r_t^2] - \mathbb{E}[r_{t+\tau}^2] \mathbb{E}[r_t^2] \\ &= \mathbb{E}[r_{t+\tau}^2 r_t^2] - \mathbb{E}[r_t^2]^2 \\ &= \mathbb{E}[\mathbb{E}[r_{t+\tau}^2 r_t^2 \mid \mathcal{F}_{t+\tau-1}]] - \mathbb{E}[r_t^2]^2 \\ &= \mathbb{E}[r_t^2 \sigma_{t+\tau}^2 \mathbb{E}[\varepsilon_{t+\tau}^2 \mid \mathcal{F}_{t+\tau-1}]] - \mathbb{E}[r_t^2]^2 \\ &= \mathbb{E}[r_t^2 (\alpha_0 + \alpha_1 r_{t+\tau-1}^2)] - \mathbb{E}[r_t^2]^2 \\ &= \alpha_0 \mathbb{E}[r_t^2] + \alpha_1 \mathbb{E}[r_{t+\tau-1}^2 r_t^2] - \mathbb{E}[r_t^2]^2 \\ &= \alpha_1 \text{Cov}(r_{t+\tau-1}^2, r_t^2) + \alpha_0 \mathbb{E}[r_t^2] + \alpha_1 \mathbb{E}[r_{t+\tau-1}^2] \mathbb{E}[r_t^2] - \mathbb{E}[r_t^2]^2 \\ &= \alpha_1 \text{Cov}(r_{t+\tau-1}^2, r_t^2) + \alpha_0 \mathbb{E}[r_t^2] - (1 - \alpha_1) \mathbb{E}[r_t^2]^2, \end{aligned}$$

Recall from the notes that

$$\mathbb{E}[r_t^2] = \text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1}.$$

By stationarity, $\mathbb{E} [r_{t+\tau-1}^2] = \mathbb{E} [r_t^2]$, so the extra terms above cancel. Therefore

$$\begin{aligned} \text{Cov} (r_{t+\tau}^2, r_t^2) &= \alpha_1 \text{Cov} (r_{t+\tau-1}^2, r_t^2) + \frac{\alpha_0^2}{1 - \alpha_1} - (1 - \alpha_1) \left(\frac{\alpha_0}{1 - \alpha_1} \right)^2 \\ &= \alpha_1 \text{Cov} (r_{t+\tau-1}^2, r_t^2) . \end{aligned}$$

Therefore by a recursive argument we have

$$\text{Cov} (r_{t+\tau}^2, r_t^2) = \alpha_1^\tau \text{Var} (r_t^2) .$$

Now by symmetry of the autocovariance and definition of correlation we have

$$\text{Corr} (r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|} .$$